

Retrieval-Augmented Generation for Knowledge-Intensive Tasks

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ABSTRACT

Large pre-trained language models store factual knowledge in their parameters, but their ability to access and manipulate knowledge is limited, leading to hallucinations. We propose a general-purpose recipe for **retrieval-augmented generation (RAG)** combining pre-trained parametric and non-parametric memory.

1. Introduction

Pre-trained neural language models cannot easily update their memory or inspect reasons for predictions. We propose RAG models where parametric memory is a seq2seq transformer and non-parametric memory is a dense vector index accessed with a neural retriever.

2. Hallucination Mitigation

By conditioning generation directly on retrieved evidence passages, RAG dramatically reduces hallucinations compared to purely parametric models and provides verifiable source attribution.

